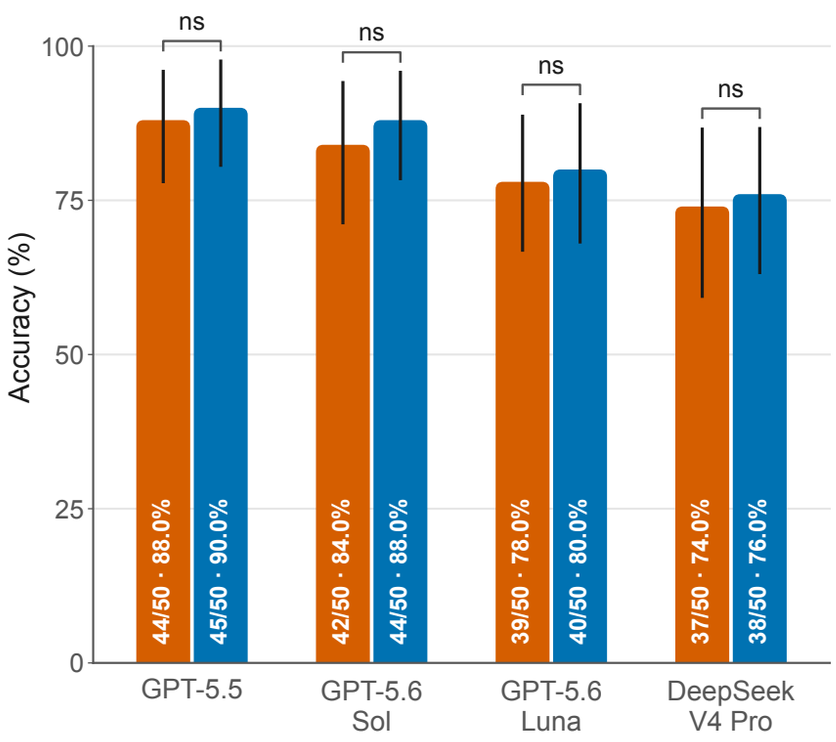
Task accuracy by model (task solved only when all 3 runs are correct)

Custom code Galaxy — 95% CI

Accuracy = tasks solved ÷ tasks (BixBench-Verified-50: 50; CompBioBench: 100; IWC: 10); a model solves a task when all 3 of its runs are correct (IWC runs: ≥ 0.99 agreement).
Bars: 95% cluster-bootstrap intervals (20,000 resamples; clusters are BixBench source capsules, otherwise tasks). Brackets: Galaxy vs custom code for each model,
paired cluster sign-flip test on tasks solved (200,000 draws; exact for IWC), Holm-adjusted over the 12 model comparisons: * adj. P < 0.05, ** < 0.01, *** < 0.001, ns not significant.

BixBench-Verified-50

50 tasks

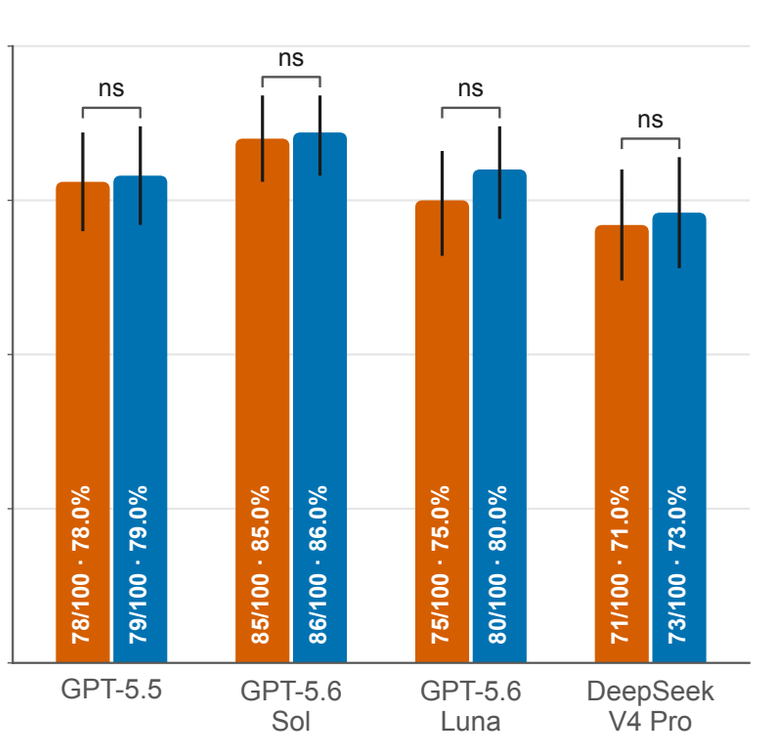


All four models: Galaxy − custom code +2.5 pts
(95% CI −4.4 to 9.9; P = 0.59, adj. over 3 benchmarks P = 0.60)

Galaxy − custom code	Δ (pts)	P	adj. P
GPT-5.5	+2.0	1.00	1.00
GPT-5.6 Sol	+4.0	0.77	1.00
GPT-5.6 Luna	+2.0	1.00	1.00
DeepSeek V4 Pro	+2.0	1.00	1.00

CompBioBench

100 tasks

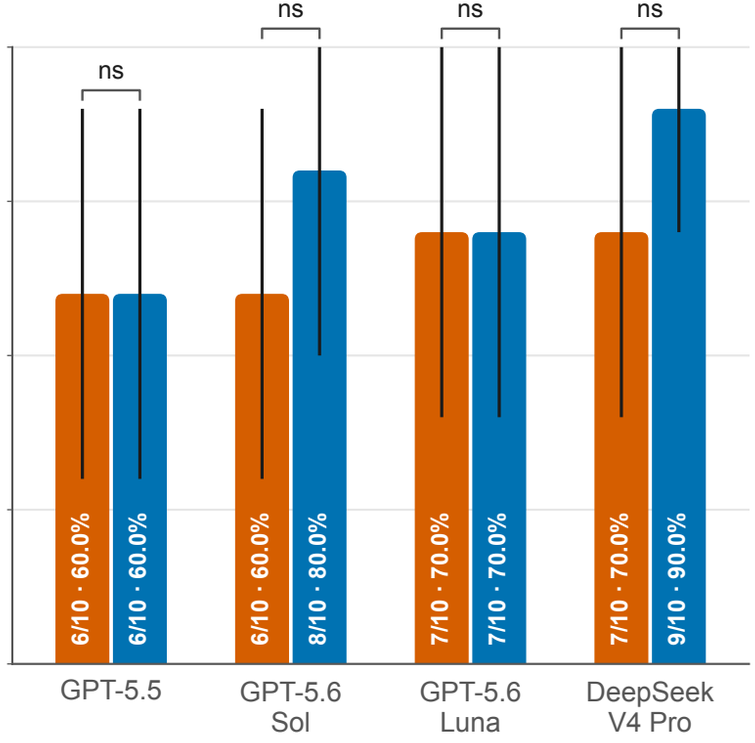


All four models: Galaxy − custom code +2.3 pts
(95% CI −1.5 to 6.0; P = 0.30, adj. over 3 benchmarks P = 0.60)

Galaxy − custom code	Δ (pts)	P	adj. P
GPT-5.5	+1.0	1.00	1.00
GPT-5.6 Sol	+1.0	1.00	1.00
GPT-5.6 Luna	+5.0	0.33	1.00
DeepSeek V4 Pro	+2.0	0.84	1.00

IWC

10 tasks; correct at ≥ 0.99 agreement



All four models: Galaxy − custom code +10.0 pts
(95% CI 2.5 to 17.5†; P = 0.13, adj. over 3 benchmarks P = 0.38)

Galaxy − custom code	Δ (pts)	P	adj. P
GPT-5.5	+0.0	1.00	1.00
GPT-5.6 Sol	+20.0	0.50	1.00
GPT-5.6 Luna	+0.0	1.00	1.00
DeepSeek V4 Pro	+20.0	0.50	1.00

† The 95% interval excludes zero but the test does not reject. With few clusters that differ between the conditions, the percentile bootstrap is too narrow; significance rests on the exact sign-flip test.